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324514 (24)

BE (5th Semester)

Examination, April-May, 2013

Branch : Electrical

CONTROL SYSTEM ENGG.

Time Allowed : Three Hours

Maximum Marks : 80

Minimum Pass Marks : 28

Note : Attempt all questions. Part (a) is compulsory and attempt any two from remaining parts. Semilog graph paper and calculators are allowed.

Q. 1. (a) Define Mason's gain formula. 2

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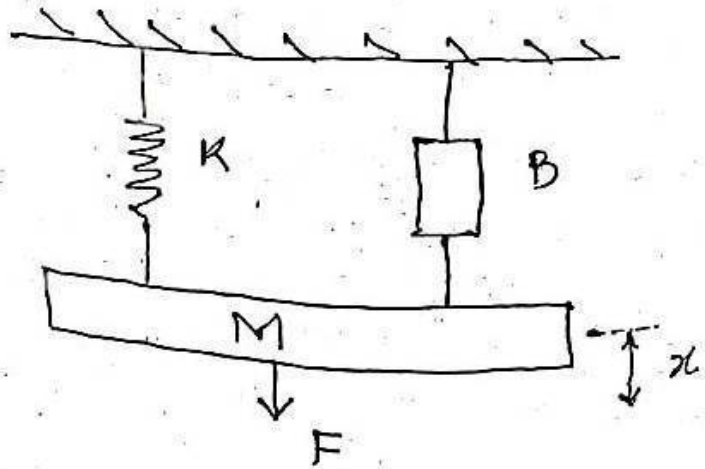
(b) Write differential equation governing the behavior of the mechanical system shown below. Obtain the analogous electric system based on force-voltage and force current analogy. Where K is spring constant B-coefficient of viscous friction, M is mass, F(t) = external force. 7



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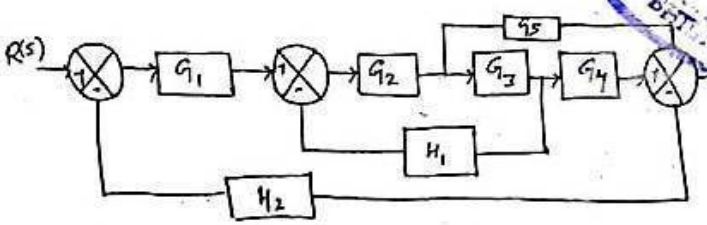
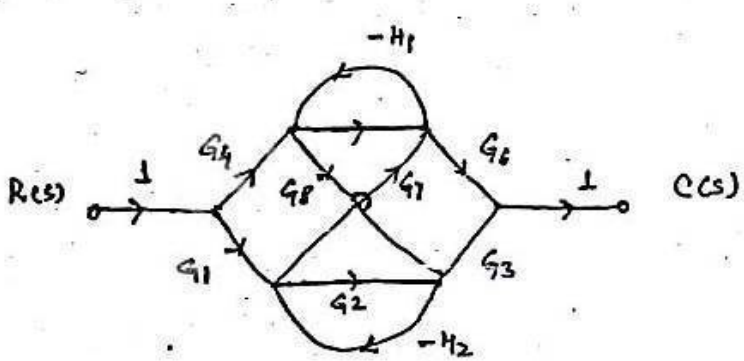
(c) Reduce the block diagram to its canonical form and obtain C(s)/R(s). 7



(d) Using Mason's gain formula, find the gain of following system. 7



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Q. 2. (a) What are standard test signals ? 2

(b) The open loop transfer function of a unity feedback system is given by :

$$G(s) = \frac{K}{s(\tau s+1)} ; \quad K>0, \ \tau>0$$

By what factor should the gain K be reduced so that the peak overshoot of unit step response of the system is reduced from 75% to 25%. 7

(c) A unit feedback control has 7

$$G(s) = \frac{K}{s(s+2)(s^2+2s+5)}$$

(i) For a unit ramp input it is desired $e_{ss} \leq 0.2$. Find K.

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(ii) Determine e_{ss} if input

$$r(t) = 2 + 4t + \frac{t^2}{2}$$

(d) The characteristic equation of feedback control system is : 7

$$s^4 + 20s^3 + 15s^2 + 2s + K = 0$$

(i) Determine the range of K for the system to be stable.

(ii) Can the system be marginally stable ? If so, find the required value of K and the frequency of sustained oscillations.

Q. 3. (a) Define minimum phase non minimum phase and all pass transfer function. 2

(b) Construct the bode plot for system whose open loop transfer function is given by : 7

$$G(s)H(s) = \frac{4}{s(1+0.5s)(1+0.08s)}$$

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Determine :

(i) Gain Margin

(ii) Phase Margin

(iii) Stability

(c) Draw the root locus for the following polynomials : 7

$$s(s + 2) + K(s + 4) = 0$$

(d) Draw the polar plot for given transfer function : 7

$$G(s)H(s) = \frac{1}{(1+ST_1)(1+ST_2)}$$

Q. 4. (a) Give the uses of compensator. 2

(b) Illustrate the advantages for use of lag-load compensator in a control system. Consider a lag load compensator defined by

$$G(s) = KC\frac{\left(s+\frac{1}{T_1}\right)\left(s+\frac{1}{T_2}\right)}{\left(s+\frac{\beta}{T_1}\right)\left(s+\frac{1}{\beta T_2}\right)}$$

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Show that at frequency $W_1 = \frac{1}{\sqrt{T_1 T_2}}$ the phase angle of $G_c(jw)$ becomes zero. 7

(c) Design a suitable lag compensator for

$$G(s) = \frac{K}{s(s+2)(s+20)}$$

to meet the following specifications : 7

(i) Phase margin > 35°

(ii) Velocity error constant ≤ 20 sec^{-1}

(d) What are components of feedback control system ? Distinguish between open loop and closed loop system. Draw necessary diagrams. 7

Q. 5. (a) Define state variable in reference to state variable analysis. 2

(b) Explain the concept of controllability and observability. If a system is controllable then it is observable also ? State whether system given ahead is controllable or not ? 7

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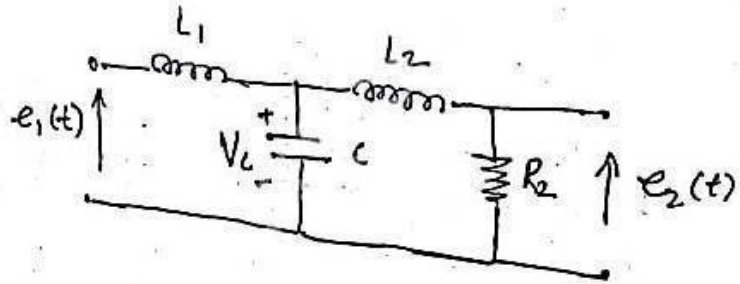
$$\begin{vmatrix} X_1 \\ X_2 \\ X_3 \end{vmatrix} = \begin{vmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -6 & 11 & -6 \end{vmatrix} \begin{vmatrix} Y_1 \\ Y_2 \\ Y_3 \end{vmatrix} + \begin{vmatrix} 0 \\ 0 \\ 1 \end{vmatrix} 4$$

(c) Obtain the state model of given electric model. 7



(d) Determine state model using signal flow graph approach of a system, whose transfer function is given 7

$$\frac{Y(s)}{U(s)} = \frac{5s+6}{s^2+2s+3}$$

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